

# UTSAV KUMAR RANJAN

Adm. No. 25MT0604

✉ u.ranjan225@gmail.com

☎ 7488637961

🌐 [linkedin.com/in/utsav-kumar-ranjan-35a584385](https://www.linkedin.com/in/utsav-kumar-ranjan-35a584385)

🐙 [github.com/firstlast](https://github.com/firstlast)



## Education

### IIT (ISM) DHANBAD

Master of Technology in Fuel, Mineral and Metallurgical Engg. (Specialization: Fuel and Energy Engg.)

Expected July 2027

Dhanbad, Jharkhand

- **Relevant Coursework:** Renewable Energy System, Advanced Thermodynamics, Powerplant, Engineering, Biofuels & Biomass Conversion, Fuel Technology, Coal and Mineral Beneficiation, CFD

### Darbhanga College of Engineering, Darbhanga

Bachelor of Technology in Mechanical Engineering (CGPA: 7.97 / 10.00)

August 2024

Darbhanga, Bihar

Year: [2020-2024]

- **Relevant Coursework:** Engineering Thermodynamics, Heat Transfer, Fluid Mechanics, Strength of Materials, Theory of Machines, Design of Machine Elements, Powerplant Engineering, Refrigeration and Air Conditioning, Manufacturing Processes

## Projects

### Comparative analysis of Rectangular and Elliptical fins in ANSYS (B.Tech) |

- Elliptical fin shows slightly better heat transfer performance than rectangular fins.
- The average surface heat transfer coefficient for an elliptical Fin is Slightly more than that of Rectangular Fin.
- The performance of elliptical fin can be further enhanced by refining their shape or profile.

## Technical Skills

**Software & Tools:** ANSYS, MS EXCEL, MS POWERPOINT

**Core Competencies:** Thermodynamics, Engineering Mechanics, Heat Transfer, Fluid Mechanics, Strength of Materials, Powerplant Engineering, Theory of Machine, Design of machine elements.

**Simulation and Modelling:** Heat Transfer and Fluid Flow Simulation

## Research Interest

- Life Cycle Assessment (LCA) and Sustainability Assessment
- Biomass and Coal Gasification
- Pyrolysis and Thermal Conversion
- Hydrogen Production, Storage and Utilization
- Waste-to-Energy and Alternative Fuels
- Sustainable Energy Generation and Decarbonization

## Achievements

- Qualified GATE(2024,2025)
- Qualified JEE
- Completed certifications (NPTEL on Mechanical vibrations and Sustainable Development)
- Represented college in sports events.

## Social Engagements and Interest

**Sports-Engagements:** Badminton, chess, cricket, table-tennis, Piano, Music, Reading, etc.